

## Final Exam R2

1. Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  and  $v \in \mathbb{R}^n$ . If  $f$  is differentiable at  $x \in \mathbb{R}^n$ , prove that the directional derivative is  $Df_v(x) = \nabla f(x) \cdot v$ . **1p**
2. Consider the curve defined by the equation  $f(x, y) = -3$ , where  $f(x, y) = 2x^2 - 4x + y^2 - 4y$ .
  - (a) Find the direction in which the function  $f$  increases the most at the point  $(2, 1)$ . **0.5p**
  - (b) Write a gradient descent method to find the minimum of  $f$ , starting from  $(2, 1)$ . **0.5p**
  - (c) Find the equation of the tangent line to the curve at the point  $(2, 1)$ . **1p**
3. Consider the function  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x, y, z) = x^3 + y^2z - xz^2 + y^3 - z^3 + x^2 - y^2$ .
  - (a) How many critical points does the function  $f$  have? **1p**
  - (b) Classify the critical point  $(0, 0, 0)$ . **1p**
  - (c) Write the second order Taylor polynomial for  $f$  centered around  $(0, 0, 0)$ . **1p**
4. Let  $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \end{bmatrix}$ ,  $b = \begin{bmatrix} 8 \\ 28 \end{bmatrix}$ ,  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ ,  $\alpha > 0$ . Let  $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ ,  $f(x) = \|x\|^2 + \alpha \|Ax - b\|^2$ .
  - (a) Find the unique  $x^* \in \mathbb{R}^3$  that solves the problem:  $\min_{x \in \mathbb{R}^3} \|x\|^2$  subject to  $Ax = b$ . **1p**
  - (b) Find the unique  $x_\alpha^* \in \mathbb{R}^3$  that minimizes  $f$  over  $\mathbb{R}^3$ . Show that  $\lim_{\alpha \rightarrow \infty} x_\alpha^* = x^*$ . **1p**
5. Compute  $\int_0^2 \int_{x^2}^4 x e^{-y^2} dy dx$  by changing the order of integration. **2p**

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Time: 2h. Total number of points: 10p. Please write your full name, your group and your row number R2.